KING FAHD UNIVERSITY OF PETROLEUM & MINERALS
UNDERGRADUATE RESEARCH OFFICE
TERM 252

Title: *Robotic Hand for Natural Speech to American-Sign Language Translation*

Members:

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Section 1
Introduction

1.1 Project Objectives:

This project aims to develop a robotic system that implements two human-like arms that translate natural speech to American Sign Language (ASL). Minimally viable communication is the current goal for this project, with basic numerical and alphabetical translation the first fundamental step. ASL is chosen due to its simplicity and widespread use with its own grammar and structure, and conveying it through human-like robotic arms allows for more accurate, expressive, and intuitive signing compared to text-based or screen-dependent alternatives.

1.2 Project Impact:

This project is impactful because it directly addresses communication barriers faced by Deaf and hard-of-hearing individuals by enabling more inclusive, real-time interaction between hearing and non-hearing communities through accessible robotics. By translating natural speech into American Sign Language (ASL), the system supports a primary and culturally significant language used by a large portion of the Deaf community in the United States and globally, making information more readily accessible in everyday environments such as public services, education, and healthcare.

1.3 Project Management:

Name	ID	Distinguished Tasks	Common Tasks
Hashem Altujar	202432940	Simulation Jetson Nano Software	Documentation
Abdullah Omar Siddiqui	202453040	Robotic Hand Assembly	
Abdulrahman	202450780	Robotic Hand Assembly	

Omar Siddiqui			
Ali Naeem Al Shaikh Ahmed	202179050	Simulation Pipeline Software Development	

Figure 1: *Project Management and Tasks Distribution*

1.4 Market & Engineering Requirements:

Engineering	Marketing
Economy: <3000 SAR in Cost	**Target Market:** Educational institutions, research labs, and hobbyists are interested in affordable AI-enabled robotics.
Weight: Lightweight (< 5kg)	**Usability:** Easy to install, operate, and maintain with minimal technical expertise required.
Performance: Fast Response; Accurate with a high degree of freedom (90% or higher); clearly understandable	**Market Differentiation:** Offers dual-arm manipulation and AI processing (Jetson Nano) at a lower cost and weight compared to similar systems.

Figure 2: *Overview of Requirements (Engineering and Marketing)*

1.5 Performance Metrics:

- **Accuracy Score:** $Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$
- **Precision:** $Precision = \frac{TP}{TP + FP}$
- **Recall:** $Recall = \frac{TP}{TP + FN}$
- **F1 Score:** $F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$

Section 2
Pipeline

2.1 Audio Capture

- Continuous audio stream with buffering (so you can capture the start of speech).
- Add either:
    - Push-to-talk (simplest and most reliable), or
    - Wake word (hands-free, but more work + more false triggers).

MVP recommendation: push-to-talk + a hardcoded "STOP" keyword always active.

2.2 Speech-to-text (STT)

Convert speech → text on Jetson.

Methodology choice:

- Start with a constrained vocabulary (letters and numbers) to raise accuracy.
- Use a confidence threshold: if confidence is low, ask for repetition (or display what was heard).

Output example: `"A 2 B 9"`

2.3 Text normalization

Normalize whatever STT gives into a clean token sequence.

- "double u" → `W`
- "oh" / "zero" → `0` (define your rule)
- "one two three" → `1 2 3`
- Remove filler words ("please", "uh", etc.)

Output: standardized tokens like `[A, 2, B, 9]`

This stage is what makes the MVP feel reliable.

2.4 ASL Mapping

For MVP, you do direct token → sign mapping:

- Alphabet (A–Z): ASL fingerspelling handshapes
- Numbers (0–9, maybe 10–20): number handshapes (define supported set)

Store each sign as a parameterized pose:

- Joint angles for each arm + hand/fingers (or servo values), plus optional wrist orientation.
- Include timing (hold duration, transition duration).

Data structure idea (conceptual):

- `SignPose`: left_arm_joints, right_arm_joints, duration, hold_time
- `SignSequence`: list of SignPose + transitions

Two-arm decision for MVP:

- Many fingerspelling signs can be done with one dominant hand while the other stays neutral.
- Use two arms mainly for:
    - symmetry / human-like presentation
    - stabilization/expressiveness later
    - future two-handed signs

So for MVP, pick:

- Right arm signs, left arm neutral (or mirrored presentation if you prefer).

2.5 Sign synthesis

This is where robotics quality matters.

For each token:

1. Move from current pose → target sign pose with a smooth trajectory
2. Hold pose (so it's readable)
3. Transition to next sign pose

Key requirements:

- Enforce joint limits, speed limits, and "comfort" limits
- Use smoothing to avoid jerky movements
- Standardize timing so the signing is legible (humans need holds)

MVP timing guideline:

- transition: ~0.3–0.8 s
- hold: ~0.3–0.7 s

  Tune after user testing.

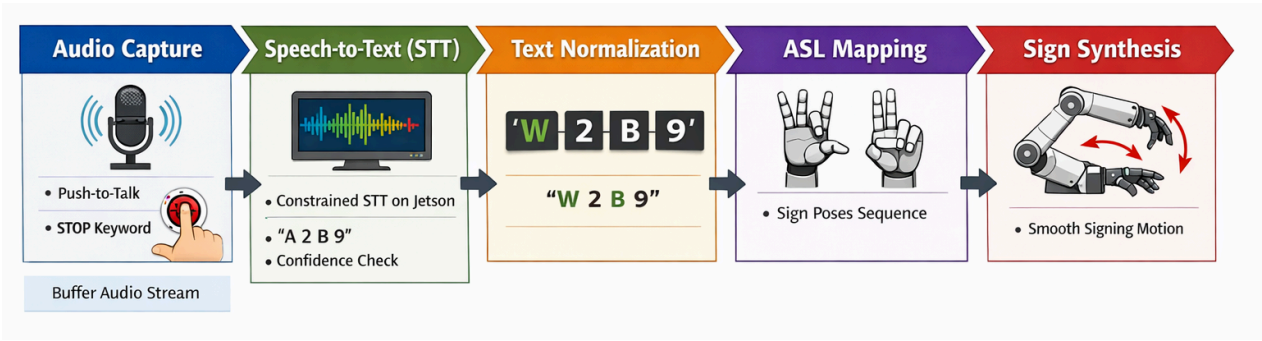

Figure 3: *Pipeline representing natural speech conversion to robotic hand sign synthesis*